

Exercise 3

Exercise 3.1 Sequential quadratic programming

In the following the SQP method shall be used to solve the *constrained nonlinear optimization problem*

$$\min_{\mathbf{x} \in \mathbb{R}^2} 5x_1^5 + 25x_1x_2 + 10x_2^3 \quad (1a)$$

$$\text{s.t. } x_1x_2^3 - 2 \leq 0 \quad (1b)$$

$$-x_1 - 2x_1x_2 + 2 \leq 0 \quad (1c)$$

$$x_1 \geq 0. \quad (1d)$$

- a) State the Lagrangian function for problem (1) and use it to approximate it at the iteration point $\mathbf{x}^k = [1, 1]^T$, $\boldsymbol{\mu}^k = [0, 0, 0]^T$ by its linear-quadratic approximation

$$\min_{\mathbf{s} \in \mathbb{R}^n} \frac{1}{2} \mathbf{s}^T \nabla_{\mathbf{x}\mathbf{x}}^2 L(\mathbf{x}^k, \boldsymbol{\lambda}^k, \boldsymbol{\mu}^k) \mathbf{s} + \nabla f(\mathbf{x}^k)^T \mathbf{s} \quad (2a)$$

$$\text{s.t. } \nabla g_i(\mathbf{x}^k)^T \mathbf{s} + g_i(\mathbf{x}^k) = 0, \quad i = 1, \dots, p \quad (2b)$$

$$\nabla h_i(\mathbf{x}^k)^T \mathbf{s} + h_i(\mathbf{x}^k) \leq 0, \quad i = 1, \dots, q. \quad (2c)$$

- b) Add-On: Draw the admissible set of the linear-quadratic problem (2) into Figure 1 and use it to determine the solution of the approximation graphically.
- c) Perform one step of the SQP method to compute the new iteration point $\mathbf{x}^{k+1}$. Is a full step size possible? How can the nonlinear constraints be considered in the determination of the step size?

Exercise 3.2 Exact penalty function

Given is the constrained optimization problem

$$\min_{x \in \mathbb{R}} x^2 \quad (3a)$$

$$\text{s.t. } x - 1 = 0, \quad (3b)$$

for which the difference between the l_1 - and the l_2 -penalty function is to be illustrated.

- a) Determine the solution to the original constrained problem using the KKT conditions.
- b) Equality-constrained problems can be solved by adding the constraint as a penalty term to the cost function and then solving the resulting unconstrained problem. Investigate the two penalty functions

$$\Psi_{l_1} = f(x) + \sigma |g(x)| = x^2 + \sigma |x - 1| \quad (4)$$

$$\Psi_{l_2} = f(x) + \sigma g(x)^2 = x^2 + \sigma (x - 1)^2. \quad (5)$$

Which condition must be satisfied for the penalty parameter σ such that the respective penalty function has a unique minimum at the optimal point of the original problem? For which of the two penalty functions can we speak of an exact penalty function in this context?

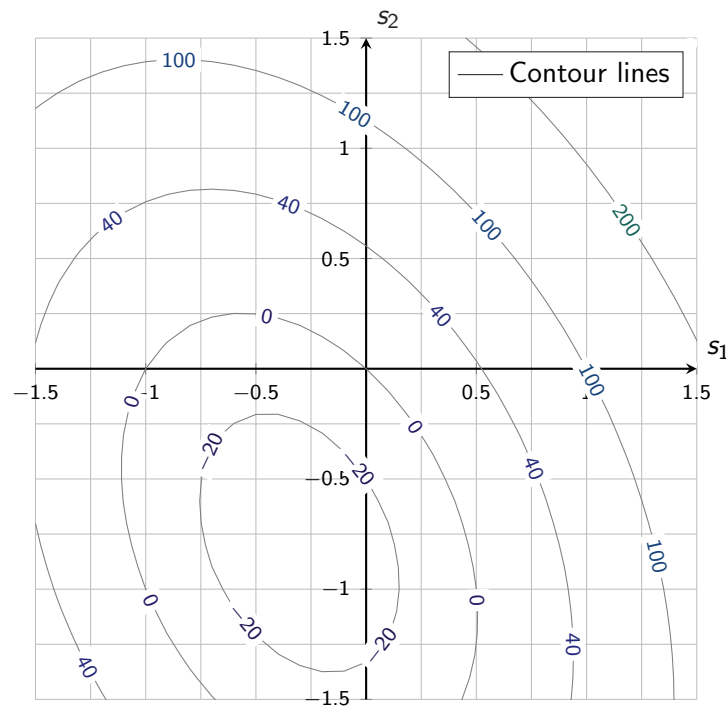


Figure 1: Contour lines of the linear-quadratic approximation of Task 1.

Exercise 3.3 Lagrangian duality

Given is the linear constrained optimization problem

$$\min_{x \in \mathbb{R}^4} c^T x, \quad c^T = [1, 2, 3, 4] \quad (6a)$$

$$\text{s.t. } Ax = b, \quad A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & -3 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ \frac{1}{2} \end{bmatrix} \quad (6b)$$

$$x \geq 0, \quad (6c)$$

which is to be solved by using the dual problem.

- State the Lagrangian function and the corresponding dual problem for the optimization problem above. How many optimization variables and constraints result for the dual problem?
- Add-On: Determine the solution to the dual problem graphically by drawing the contour lines and constraints.
- Calculate the primal solution using the obtained dual solution. Does a duality gap exist?